

Fig. 10: Equivalent circuit of a via

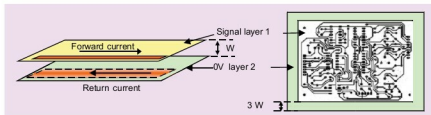


Fig. 11: Dimensions of 0V plane

between the guard trace and the trace to be protected follows 3W rule. Guard trace is connected to ground at both ends. If it is long, it should be connected to the ground plane at multiple points $\lambda/20$ apart, where λ is the minimum wavelength of the signal/clock.

Shunt trace

This is a trace located immediately below the critical clock trace. It need not be connected to ground. Best isolation can be obtained by providing both guard and shunt traces, the former providing isolation by terminating magnetic field lines (shown as dotted lines in Fig. 9) and the latter providing isolation by terminating electric field lines.

Image planes

An image plane is a plane physically adjacent (immediately below) to the signal routing plane. It provides a low-impedance path for RF currents and reduces emissions since RF currents use the plane instead of air. Power supply and ground planes normally behave as image planes, although separate image planes can also be provided.

Via inductance

Each via in a board introduces additional inductance of 1 nH and capacitance up to 0.5 pF . Fig. 10

shows an equivalent circuit of a via; it consists of a series inductor and a shunt capacitor and, hence, acts as a low-pass filter. It can cause signal delays and affect the high-frequency performance of a board. Hence, vias should be reduced to a minimum.

Power and ground planes

A continuous sheet or foil of copper for ground (and power) can provide an infinite number of parallel paths and, hence, provide least-possible inductance. At higher frequency, skin effect also starts to play its role, causing an increase in resistive part of impedance.

A plane can provide a very large surface area and least-possible resistance. A ground plane therefore is the single-most cost-effective solution to reduce ground path impedance and solve most EMC problems.

Functions of a ground plane

Ground plane serves the following EMC and signal-integrity functions:

1. It has very low RF impedance and, hence, can provide devices and circuits with stable voltages at RF, which can be distributed evenly to a large number of devices and components by avoiding voltage drops.

2. RF return currents flow via 0V reference and, if a plane is used, return current can take whichever path it wants to, but prefers a path

with lowest overall impedance (that is, one with least inductance and highest capacitance). So when a trace is routed over a ground plane, return currents will flow back right underneath the trace, reducing loop area enclosed by forward and return paths, thus reducing differential mode emission and susceptibility.

3. A plane underneath a track carrying high-frequency currents or a noisy IC will also behave as an image plane. As such fields emanating from the track or IC will be capacitively coupled to the image of the track in the plane rather than spreading out into space. This will reduce common-mode emissions.

4. Capacitive filters for blocking common-mode voltages are normally mounted between tracks entering the board and 0V reference. These bypass common-mode currents to ground, and a ground plane provides the ideal low-impedance path for these currents to ground.

Size of 0V plane

Size of a 0V plane must be such that it does not only lie beneath all components and traces of a circuit but also extends beyond these on all sides of the PCB by as much an amount as possible. This is because, although forward currents through a trace are constrained to flow in the trace, return currents flowing through a plane beneath the trace can spread out in the plane on either side of the trace. So edge of the ground plane should extend more than three times the trace-to-plane spacing (or three times the track width).

Openings and discontinuities in ground plane

Ground planes, as far as possible, should be continuous throughout. In a practical scenario, this is virtually impossible since anti-pads have to be provided in the ground plane. This causes the ground plane to be perforated, thus increasing its